



Statistical AI Agent for Dataset Analysis

Week 1 – Setup, Data Analysis, and Feature Extraction

Installation and Setup

- ☐ Install the latest version of Python.
- ☐ Install required libraries: `pandas` , `numpy` , `scikit-learn` , `matplotlib` , `seaborn` , and any other necessary tools.
- ☐ Familiarize yourself with the dataset to be used (collect or construct OK/KO datasets).
- ☐ Define the standard input file format (CSV, Excel, etc.).

Statistical Data Analysis

- ☐ Clean the data (remove missing or invalid values).
- ☐ Calculate statistical indices (mean, median, mode, variance, standard deviation, etc.) for each group: OK and KO.
- ☐ Plot histograms, boxplots, and simple graphs for an initial comparison.

Testing and Feature Review

- ☐ Check which indices or features show the greatest distinction between OK and KO groups.
- ☐ Implement a simple script for scoring or ranking distinguishing features.

- ☐ Record the initial results and document them.
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Week 2 – Machine Learning Model, Chatbot, and User Interface

Machine Learning Model (Optional)

- ☐ Choose a simple model (such as Logistic Regression or Decision Tree).
- ☐ Train the model on OK/KO data.
- ☐ Extract feature importance or model coefficients.
- ☐ Test the model and compute classification accuracy.

AI Agent and Chatbot

- ☐ Design Python functions to handle user requests (e.g., “Calculate the mean for the OK group”).
- ☐ Implement an initial chatbot capable of understanding basic commands and returning statistical outputs or charts.
- ☐ *(Optional)* Set up a local LLM (like Llama) for interpreting more complex text commands.
- ☐ Test interaction and review response quality.

GUI Development

- ☐ Select a suitable framework (e.g., Streamlit, Tkinter, or PyQt).
 - ☐ Implement a data upload form (input file).
 - ☐ Display statistical results as tables and graphs.
 - ☐ Add a chat button/box for receiving commands and showing outputs.
 - ☐ Test the integration of all parts (data → analysis → chat response → GUI display).
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Week 3 – Final Testing, Documentation, and Presentation

Final Testing and Debugging

- ☐ Fully test the program on various datasets.
- ☐ Debug and optimize the code.
- ☐ Save output reports and compare results with manual/human analysis.

Documentation and README

- ☐ Write a complete README (installation, execution, input data format, and user guide).
- ☐ Add documentation to the code (comments, function explanations, architectural overview).
- ☐ Prepare project documentation, including system architecture, rationale for method choices, experiment results, chatbot conversation samples, and limitations.

Presentation Slides Preparation

- ☐ Create presentation slides (introduction, project steps, demo and results, summary, improvement suggestions).
- ☐ Practice a 15-minute presentation, including answering possible questions.

Final Review and File Preparation

- ☐ Final review of all files (code, README, documentation, slides).
- ☐ Prepare the submission package (zip the project folder).
- ☐ Create a backup of the entire project.